Kaarthikeyan R

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.50
VELAMMAL BODHI CAMPUS	Class XII (CBSE)	2023	93.60%
RMK PATASHAALA	Class X (CBSE)	2021	93.80%

SKILLS

Languages: C++, Java, JavaScript

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems

Web Development: React.js, Node.js, Express.js, REST APIs, HTML, CSS, JWT Authentication

Database: MongoDB

Tools: Git, GitHub, VS Code

PROJECTS

Memory Allocator | GitHub | Tech Stack: C++

- Implemented a **custom dynamic memory allocator** using a fixed-size memory pool and free-list based memory management.

- Implemented **block splitting and coalescing** to efficiently manage free memory and reduce fragmentation.

- Added **memory alignment** and custom allocation/deallocation for variable-sized requests.

Bank Management System | GitHub

Tech Stack: Node.js, Express.js, MongoDB, Mongoose

- Built a **RESTful banking ledger system** using **Node.js, Express.js, MongoDB and Mongoose**, implementing user authentication, account management and transaction workflows.

- Implemented **idempotent money transfers** using unique idempotency keys and **MongoDB transactions** to ensure atomic debit/credit operations.

- Designed an **immutable double-entry ledger** with **CREDIT/DEBIT** entries and MongoDB aggregation-based balance calculation.

- Implemented **JWT authentication, bcrypt password hashing, token blacklisting with TTL-based expiry, account status validation and automated email notifications**.

Shell | GitHub | Tech Stack: C++

- Built a **Unix-like shell in C++** supporting an interactive REPL and built-in commands including exit, echo, type, pwd, cd.

- Implemented **executable discovery through PATH** and execution of external programs.

- Implemented **shell parsing** with single/double quotes and escape sequences.

- Implemented **stdout/stderr redirection** and append operations.

Ecommerce App Backend Clone | GitHub

Tech Stack: Node.js, Express.js, MongoDB, Mongoose, JWT, Cloudinary, Multer

- Developed a **RESTful backend** for an e-commerce platform with **user authentication, product, and order management**.
- Implemented **JWT authentication** with **access & refresh tokens** for secure login sessions.
- Built **custom async handler** functions with **APIError** and **APIResponse** for consistent error handling and responses.
- Integrated **Cloudinary and Multer** for image storage and file uploads.

Full-Stack To-Do List Application | GitHub

Tech Stack: React.js, Node.js, Express.js, MongoDB, Mongoose, Zod

- Developed a **full-stack to-do list application** with **create, read, update, and delete (CRUD)** functionalities.
- Implemented **backend using Node.js & Express.js** with **Mongoose** for database operations.
- Added **data validation using Zod** to ensure proper input handling.
- Built **frontend with React.js** for seamless user experience.
- Used **RESTful APIs** to connect frontend and backend effectively.

SCHOLASTIC ACHIEVEMENTS

Codeforces: Max Rating **1529(Specialist)** | **550+ problems solved**

Profile: Codeforces Profile

Topics: Dynamic Programming, Graph Algorithms, Greedy Algorithms, Binary Search, Trees, Number Theory

EXTRA CURRICULAR

- Member of **NSO (National Sports Organization) - Cricket** in college

- Participated in **CBSE Cluster Track Events** (Athletics, School Level)

- **Intra-School Overall Champion (3×)** for athletics and sports performance